

Soy protein may alleviate osteoarthritis symptoms.



Alternative and complementary therapeutic approaches, such as the use of a wide array of herbal, nutritional, and physical manipulations, are becoming popular for relieving symptoms of osteoarthritis (OA). The present study evaluated the efficacy of soy protein (SP) supplementation in relieving the pain and discomfort associated with OA. One hundred and thirty-five free-living individuals (64 men and 71 women) with diagnosed OA or with self-reported chronic knee joint pain not attributed to injury or rheumatoid arthritis were recruited for this double-blind, placebo-controlled, parallel design study. Study participants were assigned randomly to consume 40 g of either supplemental SP or milk-based protein (MP) daily for 3 months. Pain, knee range of motion, and overall physical activity were evaluated prior to the start of treatment and monthly thereafter. Serum levels of glycoprotein 39 (YKL-40), a marker of cartilage degradation, and insulin-like growth factor-I (IGF-I), a growth factor associated with cartilage synthesis, were assessed at baseline and at the end of the study. Overall, SP improved OA-associated symptoms such as range of motion and several factors associated with pain and quality of life in comparison to MP. However, these beneficial effects were mainly due to the effect of SP in men rather than women. Biochemical markers of cartilage metabolism further support the efficacy of SP in men as indicated by a significant increase in serum level of IGF-I and a significant decrease in serum level of YKL-40 compared to MP. This study is the first to provide evidence of possible beneficial effects of SP in the management of OA. Examining and verifying the long-term effects of SP on improving symptoms of OA, particularly in men, is warranted.